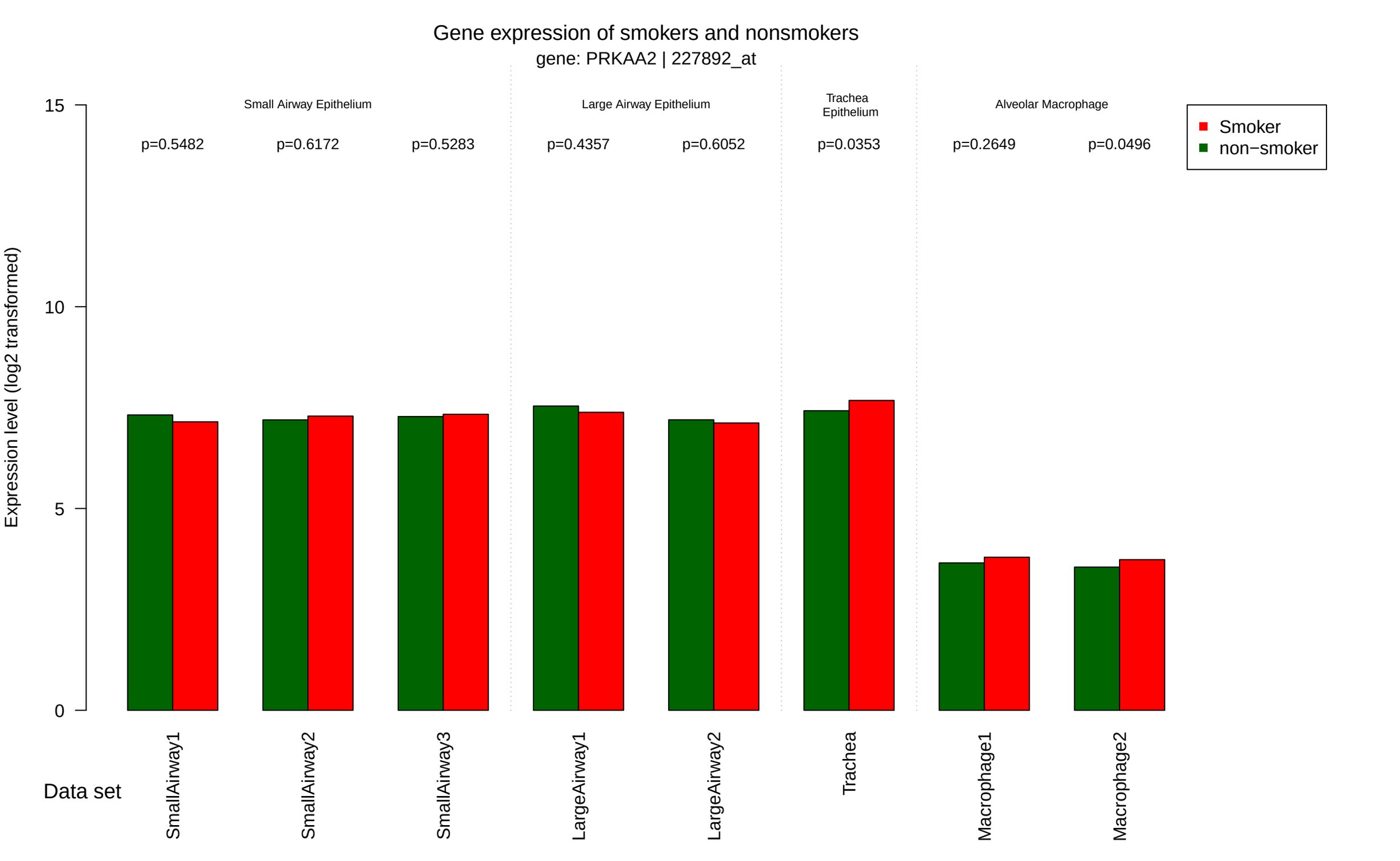
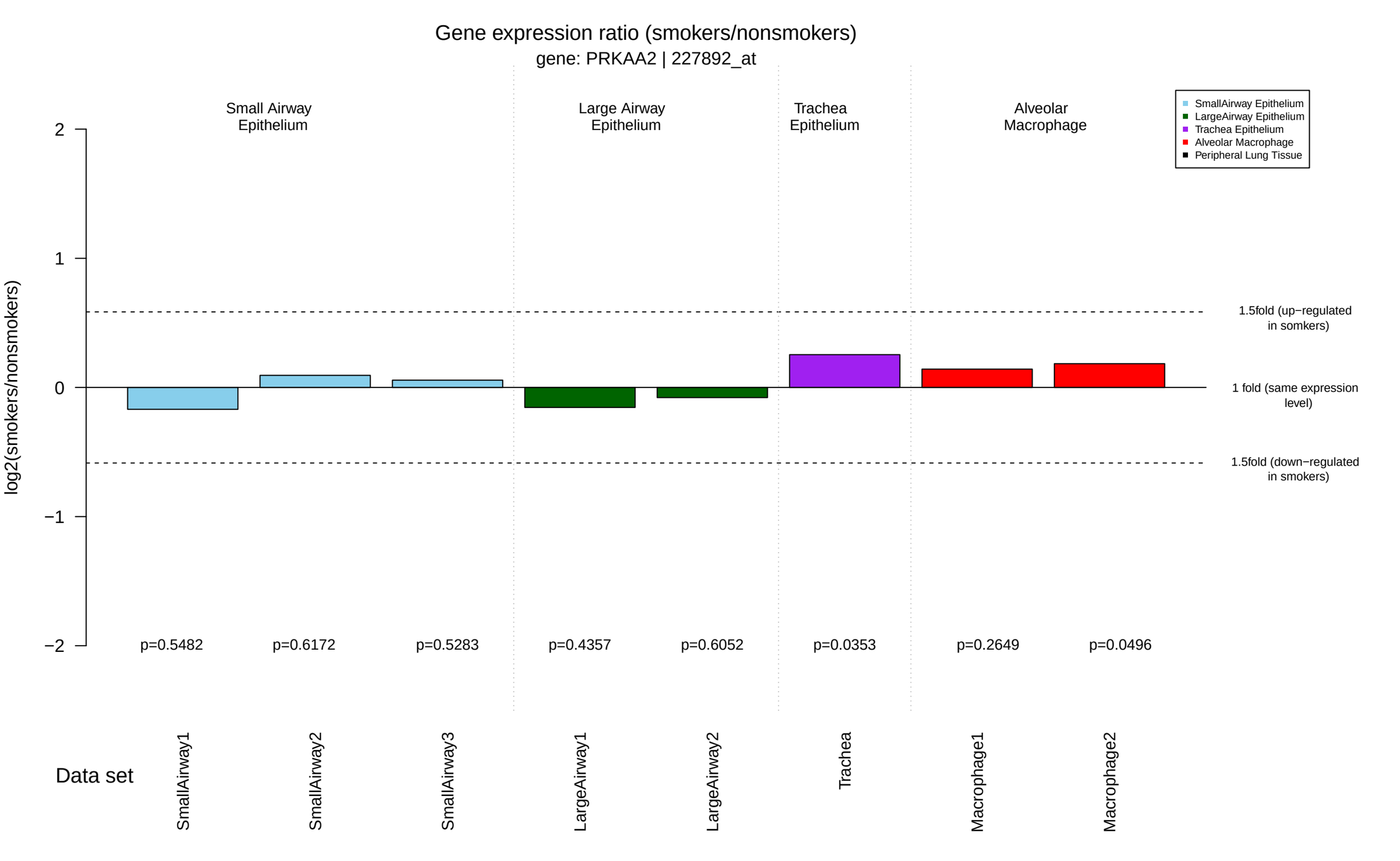
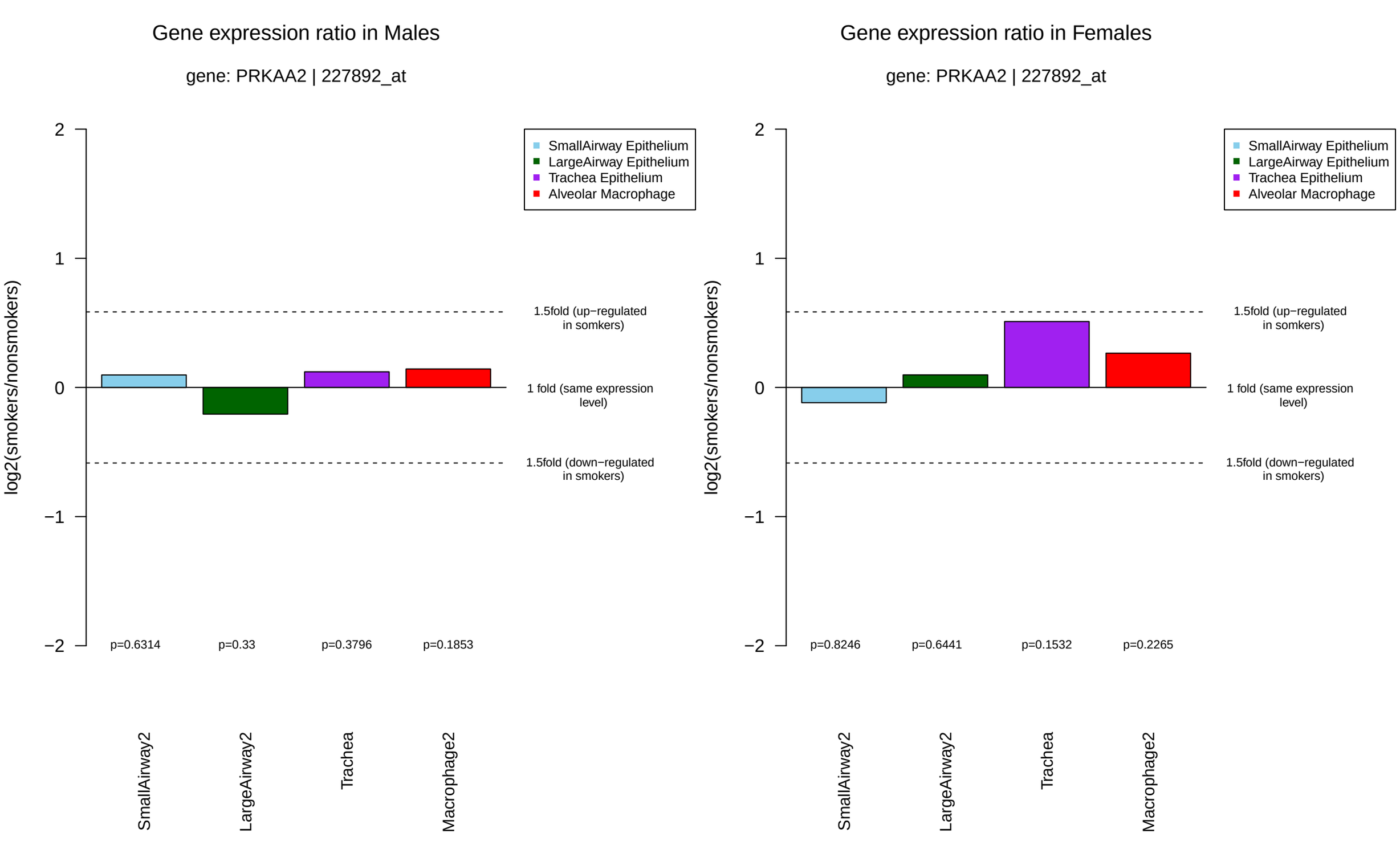
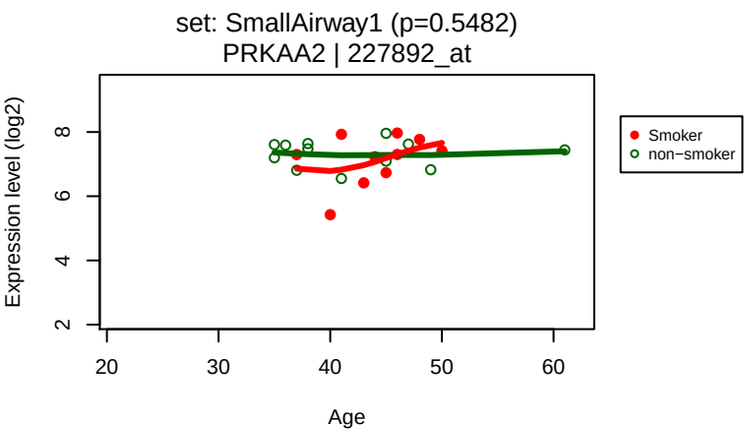
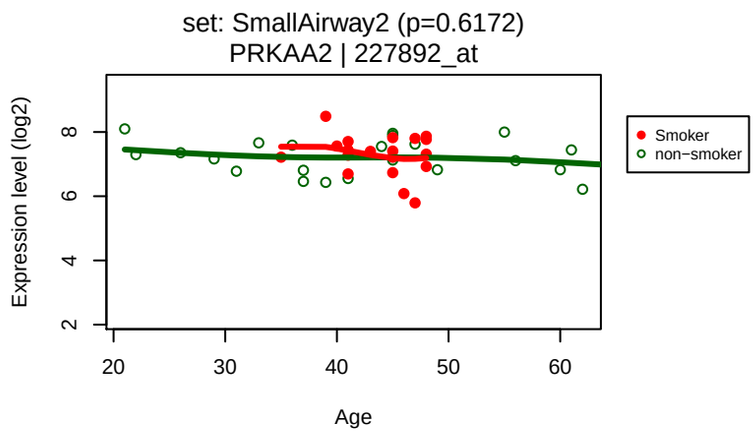
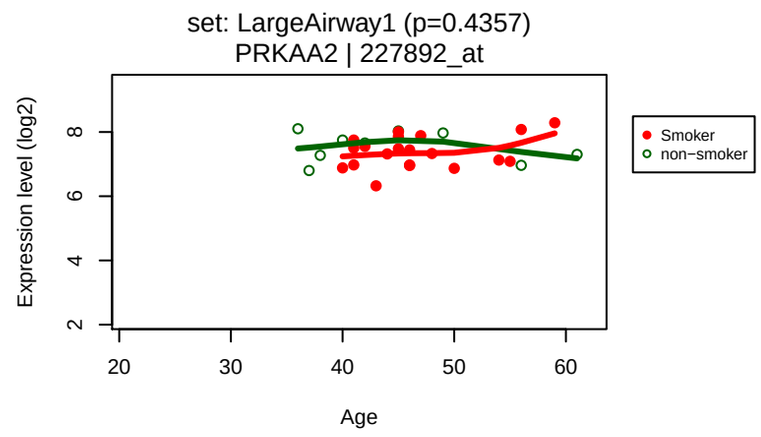

Gene expression of smokers and nonsmokers
gene: PRKAA2 | 227892_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5482
p=0.6172
p=0.5283
p=0.4357
p=0.6052
p=0.0353
p=0.2649
p=0.0496
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

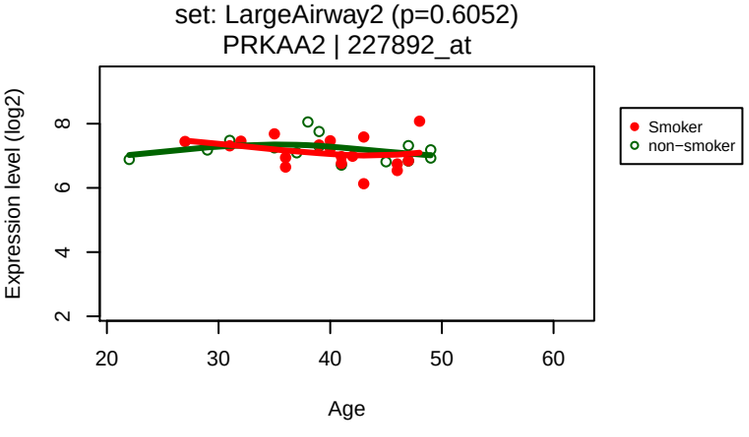

Gene expression ratio (smokers/nonsmokers)
gene: PRKAA2 | 227892_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5482
p=0.6172
p=0.5283
p=0.4357
p=0.6052
p=0.0353
p=0.2649
p=0.0496
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

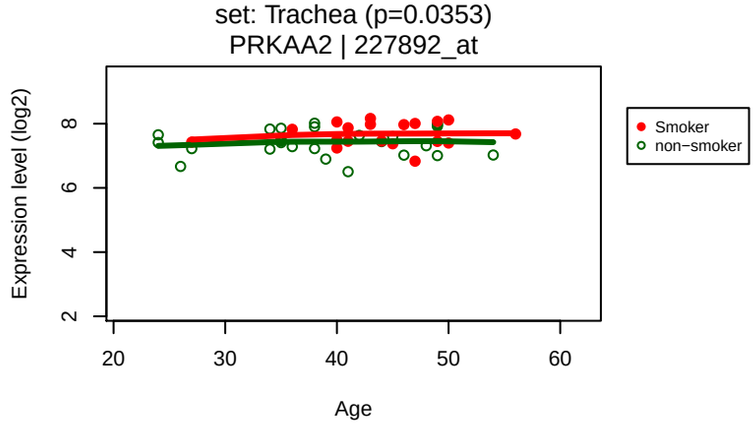

Gene expression ratio in Males
gene: PRKAA2 | 227892_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6314
p=0.33
p=0.3796
p=0.1853
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRKAA2 | 227892_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8246
p=0.6441
p=0.1532
p=0.2265
SmallAirway2
LargeAirway2
Trachea
Macrophage2

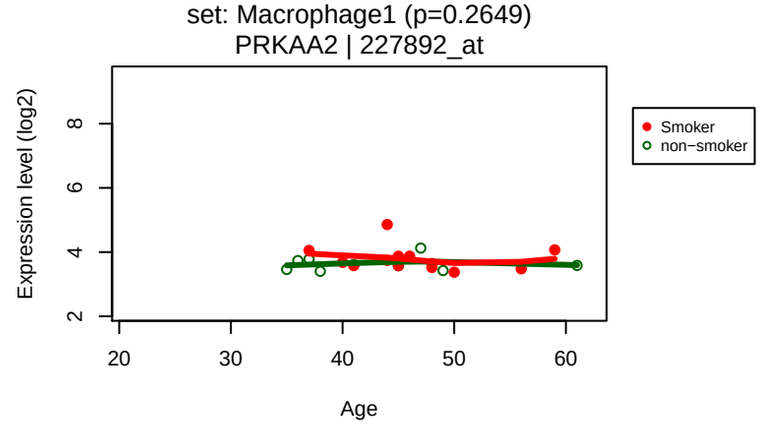

set: SmallAirway1 (p=0.5482)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker

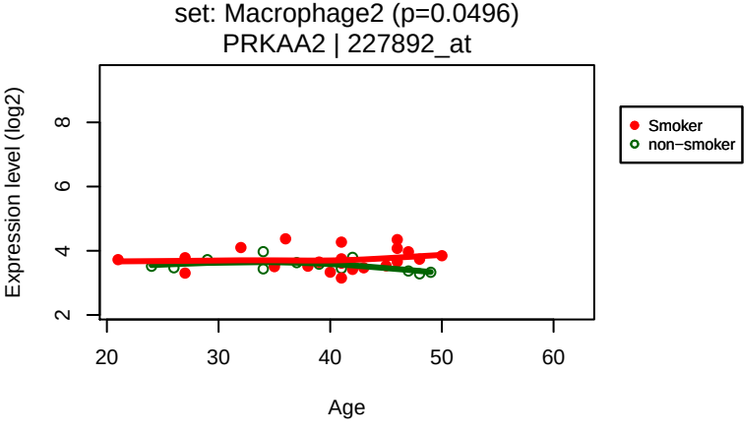

set: SmallAirway2 (p=0.6172)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker

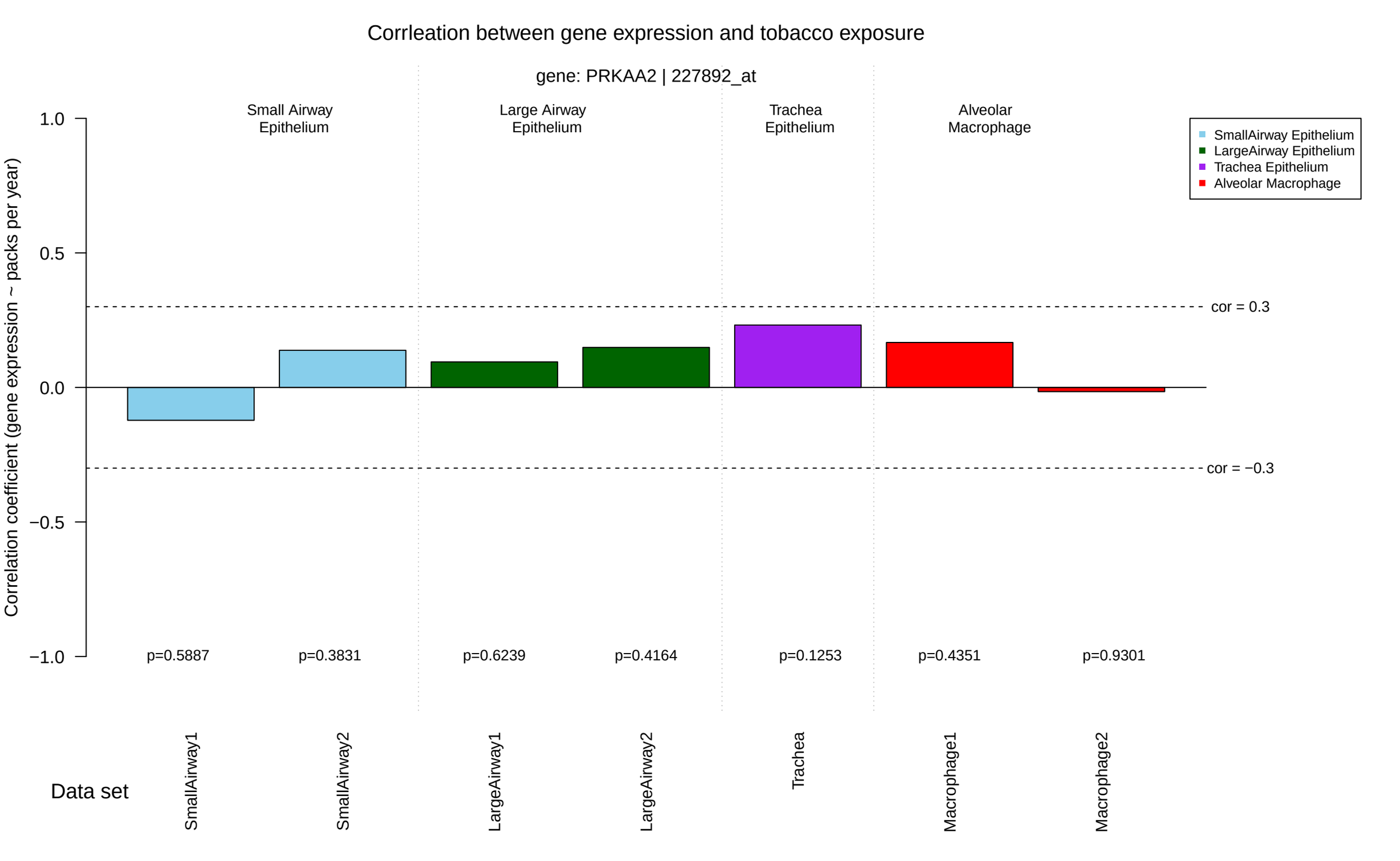

set: LargeAirway1 (p=0.4357)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6052)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0353)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2649)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0496)
PRKAA2 | 227892_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRKAA2 | 227892_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5887
p=0.3831
p=0.6239
p=0.4164
p=0.1253
p=0.4351
p=0.9301
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2